Conjectures for Primitive Pythagorean Triples

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1 Overview

We discuss the set $\mathcal{H}$ of primitive Pythagorean hypotenuses – the "hypotenuse" values that arise from primitive Pythagorean triples (right triangles with integer sides sharing no common factor) – recall the classical theorems describing its structure, and state an open conjecture about the cumulative parity of its prime elements.

A related set, $\mathcal{H}_p$, is the subset of $\mathcal{H}$ that are primes. That is, primitive Pythagorean hypotenuses that are prime.

Two of the three observations below turn out to be classical theorems — we recall them for orientation — while the third remains an open conjecture of mine, which is the focus of this note.

1. **Partition of $\mathcal{H}$** (theorem): $\mathcal{H}$ is *partitioned* into two disjoint infinite sets: the set of all powers of Pythagorean primes, and the set of all hypotenuses arising from more than one primitive Pythagorean triple.
2. **Infinitude of $\mathcal{H}_p$** (Dirichlet): The set of Pythagorean primes is infinite.
3. $\mathcal{H}_p \sim \mathbb{P}$ (conjecture): $\mathcal{H}_p$ is *similar* to and as *complex* as the primes when measured by certain cumulative-parity metrics.

2 Definitions

After a few definitions we describe a conjecture regarding Primitive Pythagorean Triples. We use standard notation, associating the natural numbers with the symbol $\mathbb{N}$[1] and we use the symbol $\mathbb{P}$ to denote the set of prime numbers.

Definition 1. *Pythagorean Triples*, denoted $\mathcal{T}$, are triples of numbers representing the lengths of the sides of right triangles where all the sides are integers.[2] Specifically,

$$\mathcal{T} \quad = \quad \{(x, y, z) \mid (x, y, z) \in \mathbb{N}^3 \ \wedge \ x^2 + y^2 = z^2\}. \tag{1}$$

Definition 2. *Primitive Pythagorean Triples*, denoted $\mathcal{T}_p$, are Pythagorean triples whose legs share no common factor:[3]

$$\mathcal{T}_p \quad = \quad \{(x, y, z) \mid (x, y, z) \in \mathbb{N}^3 \ \wedge \ \gcd(x, y) = 1 \ \wedge \ x^2 + y^2 = z^2\}. \tag{2}$$

Definition 3. *Primitive Pythagorean Hypotenuses*, denoted $\mathcal{H}$, are defined by:

$$\mathcal{H} \quad = \quad \{z \mid \exists (x, y) \in \mathbb{N}^2,\ (x, y, z) \in \mathcal{T}_p\}. \tag{3}$$

This is the set of all possible hypotenuses of primitive Pythagorean triples.

Definition 4. *Pythagorean Primes*, denoted $\mathcal{H}_p$, are defined by:

$$\mathcal{H}_p \quad = \quad \{z \mid \exists (x, y) \in \mathbb{N}^2,\ (x, y, z) \in \mathcal{T}_p \ \wedge \ z \in \mathbb{P}\}. \tag{4}$$

This is the set of primes that occur as hypotenuses of primitive Pythagorean triples.

Definition 5. *Duplicate Primitive Hypotenuses*, denoted $\mathcal{H}_d$, are defined by:

$$\mathcal{H}_d \quad = \quad \{z \mid \exists\, (x_1, x_2, y_1, y_2) \in \mathbb{N}^4,\ (x_1, y_1, z) \in \mathcal{T}_p \ \wedge \ (x_2, y_2, z) \in \mathcal{T}_p \ \wedge \ \{x_1, y_1\} \neq \{x_2, y_2\}\}. \tag{5}$$

This is the set of primitive hypotenuses that arise from more than one *distinct* primitive Pythagorean triple (i.e. not merely a swap of legs).

Definition 6. *Primitive Hypotenuses Powers*, denoted $\mathcal{H}_u$, are defined by:

$$\mathcal{H}_u \quad = \quad \{z \mid \exists\, (p, n) \in \mathcal{H}_p \times \mathbb{N},\ z = p^n\}. \tag{6}$$

This is the set of all powers of Pythagorean primes.

Definition 7. The *Prime Parity* function, Π, is defined on any prime p with $p \geq 5$ by:

$$\Pi(p) = \begin{cases} 1 & \exists n \in \mathbb{N}, p = 6n + 1 \\ -1 & \exists n \in \mathbb{N}, p = 6n - 1 \end{cases} \tag{7}$$

Definition 8. The *First Deviation* function, Δ_p, is defined by:[4]

$$\Delta_p(n) = \min_m \left\{ m \;\middle|\; n \leq \left(\sum_{i=1}^{m} \Pi(\mathcal{H}_p[i]) - \sum_{i=3}^{m+2} \Pi(\mathbb{P}[i]) \right) \right\} \tag{8}$$

[1] We assume that $\mathbb{N}$ does *not* include 0.

[2] Triples are taken as ordered, so $(3, 4, 5)$ and $(4, 3, 5)$ are distinct elements of $\mathcal{T}$.

[3] Since any common divisor of x and y also divides z, $\gcd(x, y) = 1$ is equivalent to $\gcd(x, y, z) = 1$, the usual primality condition.

[4] If for a given n, $\left\{ m \;\middle|\; n \leq \left(\sum_{i=1}^{m} \Pi(\mathcal{H}_p[i]) - \sum_{i=3}^{m+2} \Pi(\mathbb{P}[i]) \right) \right\} = \emptyset$, we set $\Delta_p(n) = \infty$.

Notes:

- The subscript u in the name $\mathcal{H}_u$ is not yet justified.
- $\mathcal{H}_p$ is a *proper* subset of the primes: $\mathcal{H}_p \subset \mathbb{P}$.
- If the sets above are treated as lists, they are indexed in sorted order.
- Clearly, Δ_p is a non-decreasing function.

First we list a few facts and state a couple of known theorems before proceeding:

1. The Pythagorean primes are of the form $4n+1$. This is a special case of a more general theorem proven by Pierre de Fermat, known as Fermat's theorem on sums of two squares. According to this theorem, an odd prime number p can be expressed as the sum of two squares if and only if p is congruent to 1 modulo 4, i.e., $p = 4n + 1$ for some positive integer n.
2. Moreover, the distribution of primes of the form $4n + 1$ has been studied extensively. The German mathematician Peter Gustav Lejeune Dirichlet proved that there are infinitely many primes of the form $4n + 1$, a result known as Dirichlet's theorem on arithmetic progressions. This theorem, which generalizes the case of Pythagorean primes, states that for any two positive coprime numbers a and d, there are infinitely many primes of the form $a + nd$, where n is a non-negative integer.
3. In the case of Pythagorean primes, we can take $a = 1$ and $d = 4$ to get the form $4n + 1$. Dirichlet's theorem thus guarantees that there are infinitely many Pythagorean primes. This is a profound result, as it extends the prime number theorem, which states that there are infinitely many prime numbers, to arithmetic progressions.

From these results we get the following characterization of Pythagorean primes:

Theorem 1. $\mathcal{H}_p = \{p \mid p \in \mathbb{P} \ \wedge \ p \equiv 1 \pmod 4\}$.

And the following structure theorem for Pythagorean hypotenuses:

Theorem 2. *The set $\mathcal{H}$ has a unique factorization (up to ordering) in $\mathcal{H}_p$: each element of $\mathcal{H}$ can be uniquely written as a product of powers from $\mathcal{H}_p$,*

$$\forall h \in \mathcal{H},\ \exists! N \in \mathbb{N},\ \exists!(\mathbf{p}, \mathbf{k}) \in \left(\mathcal{H}_p^N \times \mathbb{N}^N\right) : h = \prod_{i=1}^{N} p_i^{k_i}, \tag{9}$$

where uniqueness is up to permutation of the index i.

A classical consequence is the following partition theorem, which I had originally stated as a conjecture but which follows from the count of primitive representations of n as a sum of two squares:

Theorem 3. *The set $\mathcal{H}$ is partitioned by $\mathcal{H}_d$ and $\mathcal{H}_u$:*

1. $\mathcal{H} = \mathcal{H}_d \cup \mathcal{H}_u$;
2. $\emptyset = \mathcal{H}_d \cap \mathcal{H}_u$;
3. $|\mathcal{H}_d| = \infty$.

Proof sketch. For $n \in \mathcal{H}$, the number of primitive Pythagorean triples with hypotenuse n equals 2^{k-1}, where k is the number of distinct prime factors of n (all of which lie in $\mathcal{H}_p$ by Theorem 2). This count is a classical consequence of the unique factorization of n in the Gaussian integers. Hence n has exactly one primitive representation iff $k = 1$, i.e. iff n is a power of a single Pythagorean prime ($n \in \mathcal{H}_u$); otherwise $n \in \mathcal{H}_d$. This gives the partition. The infinitude of $\mathcal{H}_d$ follows from the infinitely many distinct products $p \cdot q$ with $p, q \in \mathcal{H}_p$ and $p \neq q$ (Dirichlet). $\square$

3 McIntire's Pythagorean Triple Conjectures

The remaining open conjecture concerns the *cumulative parity* of $\mathcal{H}_p$ — the running sum $\sum_{i=1}^{n} \Pi(\mathcal{H}_p[i])$ — and how it compares to the corresponding sum over all primes ≥ 5.

Conjecture 1. *The cumulative parity of $\mathcal{H}_p$ exhibits the following non-trivial behavior:*

1. $\displaystyle\lim_{n\to\infty} \frac{\sum_{i=1}^{n} \Pi(\mathcal{H}_p[i])}{n} = 0$;
2. $\forall n \in \mathbb{N}, \Delta_p(n) < \infty$;
3. $\displaystyle\lim_{n\to\infty} \frac{\Delta_p(n)}{n^{\ln(2\pi)}} = 41.$[5]

The three sub-conjectures together suggest that the Pythagorean primes are "as complex as" the full set of primes when measured by their parity behavior. Sub-conjecture 1 says the ± 1 values $\Pi(\mathcal{H}_p[i])$ average to zero, so $\mathcal{H}_p$ is spread evenly between residues 1 and $-1 \pmod 6$ in the long run, mirroring $\mathbb{P}$. Numerically, however, the cumulative parity of $\mathcal{H}_p$ tends to run lower than that of $\mathbb{P}$; sub-conjectures 2 and 3 quantify how often and how far the $\mathcal{H}_p$ sum nonetheless overtakes the $\mathbb{P}$ sum, asserting that deviations of any size n do occur ($\Delta_p(n) < \infty$) and that they grow at the polynomial rate $n^{\ln(2\pi)}$. This "mixing" is what we mean by saying $\mathcal{H}_p$ is non-trivial in a way similar to $\mathbb{P}$.

[5]The exponent $\ln(2\pi) \approx 1.838$ and the constant 41 are based on numerical fitting; we do not have a heuristic derivation. Readers should treat the precise values as empirical, subject to revision as more data becomes available.